

Local affine-isometry chart on a smaller ball

`MathlibAnnex.IsometryEquiv.exists_affineExtension_eqOn_ball`

Constructs the local ambient affine chart used in the gluing proof for open connected domains.

Statement

Suppose a set isometry $f: s \rightarrow t$ is defined on subsets containing the ambient balls $\text{ball}(c, R)$ and $\text{ball}(f(c), R)$, with $R > 0$. Then there is an ambient real affine isometry equivalence A that agrees with f at every source point lying in the smaller ball $\text{ball}(c, R/8)$.

Assumptions

Real normed vector spaces; f is an isometry equivalence s to t ; $R > 0$; radius- R balls about c and $f(c)$ are contained in s and t .

Conclusion

An ambient affine isometry equivalence agrees with f on the ball about c of radius $R/8$.

Proof idea

A radial extension centered at c is built from values of f on the sphere of radius $R/8$. Local midpoint preservation and positive homogeneity make the radial map globally midpoint-preserving; norm preservation then turns it into a real linear isometry. Its range contains an open ball, so it is surjective. Translating this linear equivalence produces the required affine chart, and the local radial formula proves agreement with f .

Proof steps

1. Set the working radius to $r=R/8$ and construct the radial map T from f .
2. Establish $T(0)=0$, preservation of norms, positive homogeneity, and local midpoint preservation; rescaling upgrades midpoint preservation to all vectors.
3. Package T first as an additive homomorphism and then as a real linear isometry.
4. Show the range of the linear isometry contains $\text{ball}(0, r)$ by pulling small target vectors back through f and applying the local radial formula.
5. A submodule with nonempty interior is the whole space, so the linear isometry is surjective and becomes a linear isometry equivalence.
6. Translate the linear equivalence from c to $f(c)$, and use the local radial formula once more to prove agreement with f on $\text{ball}(c, R/8)$.

Main citations

`_private.MathlibAnnex.Analysis.Normed.Affine.IsometryExtension.Local.0.MathlibAnnex.IsometryEquiv.midpointLinearIsometry`

The proof or construction of Local affine-isometry chart on a smaller ball uses the project declaration “Linear isometry induced by midpoint and norm preservation” at the indicated step.

`_private.MathlibAnnex.Analysis.Normed.Affine.IsometryExtension.Local.0.MathlibAnnex.IsometryEquiv.radialMap_midpoint_of_norm_lt`

The proof or construction of Local affine-isometry chart on a smaller ball uses the project declaration “Local midpoint preservation of the radial map” at the indicated step.

`_private.MathlibAnnex.Analysis.Normed.Affine.IsometryExtension.Local.0.MathlibAnnex.IsometryEquiv.map_midpoint_of_local_of_smul_pos`

The proof or construction of Local affine-isometry chart on a smaller ball uses the project declaration “Positive homogeneity globalizes local midpoint preservation” at the indicated step.

`_private.MathlibAnnex.Analysis.Normed.Affine.IsometryExtension.Local.0.MathlibAnnex.IsometryEquiv.radialMap_smul_of_pos`

The proof or construction of Local affine-isometry chart on a smaller ball uses the project declaration “Positive homogeneity of the radial map” at the indicated step.

`_private.MathlibAnnex.Analysis.Normed.Affine.IsometryExtension.Local.0.MathlibAnnex.IsometryEquiv.radialMap`

The proof or construction of Local affine-isometry chart on a smaller ball uses the project declaration “Radial extension map” at the indicated step.

`_private.MathlibAnnex.Analysis.Normed.Affine.IsometryExtension.Local.0.MathlibAnnex.IsometryEquiv.radialMap_eq_sub_of_norm_lt`

The proof or construction of Local affine-isometry chart on a smaller ball uses the project declaration “The radial map agrees locally with translated f ” at the indicated step.

`_private.MathlibAnnex.Analysis.Normed.Affine.IsometryExtension.Local.0.MathlibAnnex.IsometryEquiv.norm_radialMap`

The proof or construction of Local affine-isometry chart on a smaller ball uses the project declaration “The radial map preserves norms” at the indicated step.

Lean source signature (exact)

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theorem exists_affineExtension_eq0n_ball
  {s : Set E} {t : Set F} (f : s  $\approx_i$  t) (c : s) {R :  $\mathbb{R}$ } (hR : 0 < R)
  (hsball : ball (c : E) R  $\subseteq$  s)
  (htball : ball ((f c : t) : F) R  $\subseteq$  t) :
   $\exists A : E \approx^{a_i}[\mathbb{R}] F, \forall x : s,$ 
  (x : E)  $\in$  ball (c : E) (R / 8)  $\rightarrow A$  (x : E) = ((f x : t) : F)
```

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